

MTH 446: Lecture # 2 (catching up)

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Lecture Span

1. Properties of $|z|$

If $z = x + iy$, then $|z| = \sqrt{x^2 + y^2} = |\bar{z}|$. Some more properties:

1. $\forall z \in \mathbb{C}$, $|z| \geq 0$ and $|z| = 0 \iff z = 0$
2. $\forall z_1, z_2 \in \mathbb{C}$, $|z_1 z_2| = |z_1| |z_2|$
3. (Triangle inequality), $\forall z_1, z_2 \in \mathbb{C}$, $|z_1 + z_2| \leq |z_1| + |z_2|$

Proof. (Proof of 2).

$$|z_1 z_2|^2 = (z_1 z_2)(\overline{z_1 z_2}) \quad (1)$$

$$= (z_1 z_2)(\bar{z}_1 \bar{z}_2) \quad (2)$$

$$= (z_1 \bar{z}_1)(z_2 \bar{z}_2) \quad (3)$$

$$= |z_1|^2 |z_2|^2 \quad (4)$$

□

Proof. (Proof of 3)

$$|z_1 + z_2|^2 = (z_1 + z_2)(\overline{z_1 + z_2}) \quad (5)$$

$$= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) \quad (6)$$

$$= |z_1|^2 + z_1 \bar{z}_2 + z_2 \bar{z}_1 + |z_2|^2 \quad (7)$$

$$= |z_1|^2 + 2 \operatorname{Re}(z_1 \bar{z}_2) + |z_2|^2 \quad (8)$$

$$\leq |z_1|^2 + |z_1| |z_2| + |z_2|^2 \quad (9)$$

$$= (|z_1| + |z_2|)^2 \quad (10)$$

□

Corollary 0.1. $\forall z_1, z_2 \in \mathbb{C}$, $||z_1| - |z_2|| \leq |z_1 - z_2|$

Proof.

$$|z_1| = |z_2 + (z_1 - z_2)| \quad (11)$$

$$\leq |z_2| + |z_1 - z_2| \quad (12)$$

$$|z_1| - |z_2| \leq |z_1 - z_2| \quad (13)$$

□

You also can construct an alternative form of the equation of the circle centered at z_0 with radius R :

$$|z - z_0| = R \iff |z|^2 - 2 \operatorname{Re}(z \bar{z}_0) + |z_0|^2 = R^2 \quad (14)$$

Example: given $13 = 2^2 + 3^2$ and $74 = 5^2 + 7^2$, express $962 = 13 \cdot 74$ as the sum of two squares.

Proof. Let $z = 2 + 3i$ and $w = 5 + 7i$, then

$$|zw|^2 = |z|^2|w|^2 \quad (15)$$

$$= 13 \cdot 74 = 962 \quad (16)$$

Then calculate

$$zw = -11 + 29i \quad (17)$$

Thus,

$$962 = 11^2 + 29^2 \quad (18)$$

□

Polar coordinates

Let $z \in \mathbb{Z}$ and $z \neq 0$. Let $r = |z|$, then

$$\arg(z) = \{\theta + 2\pi n : n \in \mathbb{Z}\} \quad (19)$$

This is called the argument of z .